Regression Output Explanation and Index Definition

This merged document integrates the content of `Regression_Output_Explanation.md` and `Index_Define.md` into a single structure.
The graph explanations and images are preserved, and the corresponding country-level index definitions are placed alongside the relevant country sections so that the two documents complement each other directly.

1. Rule-of-Law Index Definition

Overview

The rule-of-law (RoL) variable used in this study is a constructed proxy index based on qualitative evidence from Ford Foundation reports.

As the Ford Foundation does not provide standardized quantitative country-year data, this study transforms qualitative descriptions of legal and governance-related activities into a structured quantitative measure.

The index captures the relative intensity of engagement in rule-of-law-related activities, including:

- Legal education
- Public interest law
- Access to justice
- Governance reform
- Human rights
- Civil society participation

The index is unitless and should be interpreted as an ordinal measure rather than a continuous variable. It reflects the intensity of engagement rather than actual funding amounts.

In the current `Regression.py` implementation, the plotted country-year variable is an integer index score. The country tables in Section 3 therefore report the exact index scores currently used in the graphs.

Construction Process

Step	Description	Operationalization
1. Identification	Extract rule-of-law-related activities from Ford Foundation reports	Legal education, governance reform, public interest law, access to justice
2. Coding	Assign country-period engagement based on observable characteristics	Presence, scope, continuity
3. Index Construction	Convert qualitative information into numerical values	Ordinal scale (0-8)

Coding Criteria

The index is constructed based on three observable dimensions:

Dimension	Definition	Observable Indicators
Presence	Whether activities exist	Any documented rule-of-law program
Scope	Breadth and diversity of activities	Number of programs, sectors covered
Continuity	Duration and persistence over time	Multi-year or sustained engagement

Index Scale Interpretation

Index Value	Criteria	Interpretation
0	No presence	No observable rule-of-law engagement
1-2	Low presence, limited scope, short duration	Isolated or small-scale activities
3-4	Moderate scope	Multiple activities across limited domains

Index Value	Criteria	Interpretation
5-6	High scope and continuity	Sustained engagement across multiple domains
7-8	High scope, high continuity, institutionalized presence	Long-term, large-scale engagement

Note: the index is ordinal and does not represent proportional differences in actual investment.

Temporal Coding

Period	Coding Principle
Pre-engagement period	Score = 0 or minimal
Initial phase	Low scores (1-3)
Expansion phase	Medium scores (3-5)
Mature phase	High scores (5-8)

Example:

Country	Period	Interpretation
China	Pre-1990	No Ford Foundation engagement
China	1990s	Initial legal reform programs
China	2000+	Expanded institutional engagement

Rule-of-Law Activities (Source-Based)

Category	Activities	Description	Source
Legal Education	Law schools, clinical programs, training	Capacity building in legal systems	Many Roads to Justice; China Law Program
Public Interest Law	Legal aid, advocacy, NGO legal work	Legal representation and social justice	Pursuit of Justice

Category	Activities	Description	Source
Access to Justice	Legal services for disadvantaged groups	Improving justice accessibility	Many Roads to Justice
Governance Reform	Judicial reform, institutional capacity	Strengthening state systems	China Law Program; Close to Home
Human Rights	Rights litigation, enforcement	Legal protection mechanisms	Close to Home
Civil Society	NGO participation, accountability	Social engagement and oversight	Pursuit of Justice

Country-Level Scoring Logic

Scores are assigned based on:

- Presence of activities
- Number and diversity of activities
- Duration of engagement

Scores reflect relative intensity, not exact annual values.

Cross-Country Index Profile Used in Current Plots

Country	1990 Index Score	2015 Index Score	Index Pattern
China	0	6	No activity pre-1990 in the conceptual framework; gradual expansion after legal reforms in the plotted series
India	2	5	Continuous but moderate expansion
Brazil	3	6	Expansion in governance and legal advocacy
Mexico	2	5	Gradual increase in institutional reform
Kenya	1	5	Increasing engagement over time
Nigeria	1	4	Limited and less continuous engagement

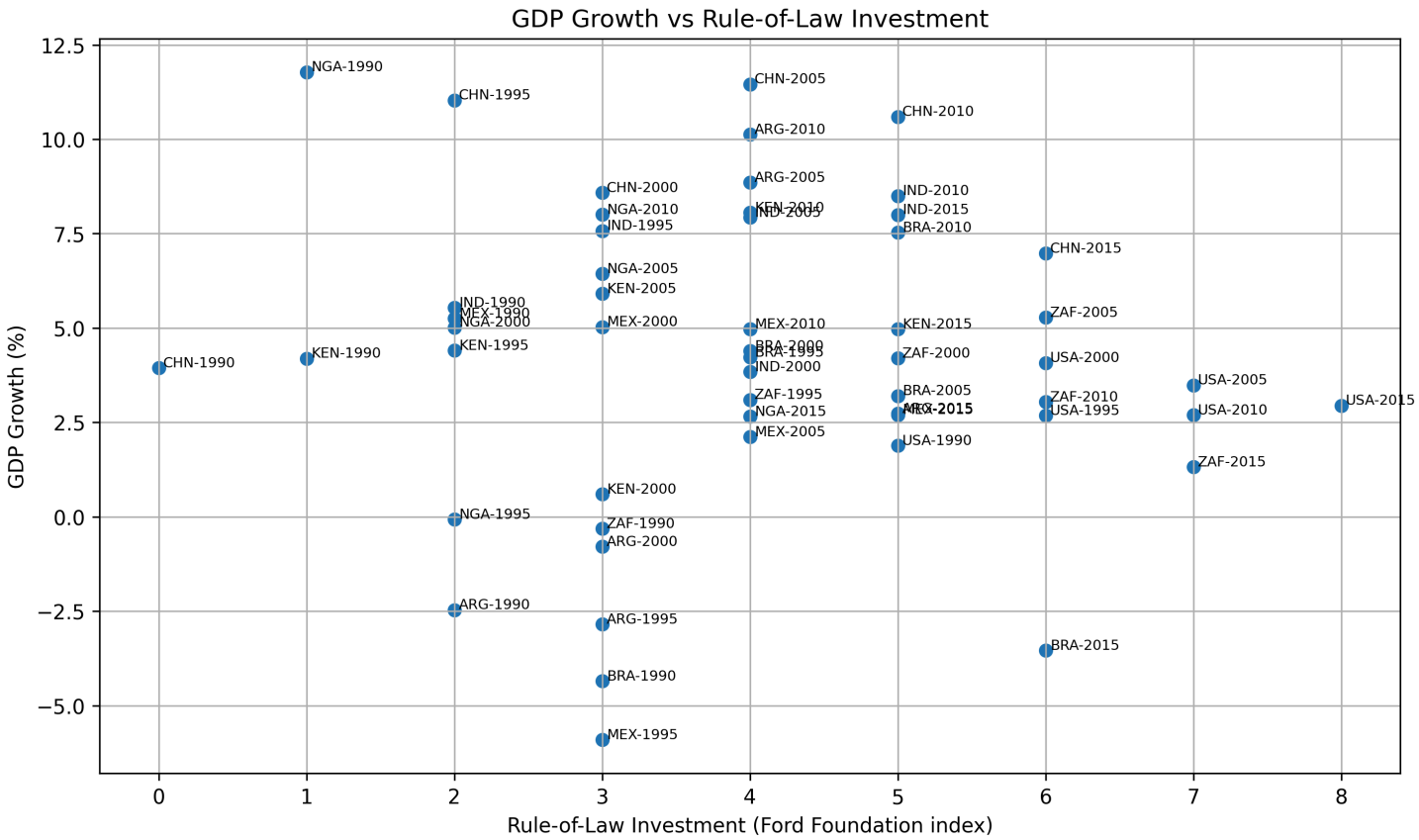
Country	1990 Index Score	2015 Index Score	Index Pattern
South Africa	3	7	Strong post-1990 institutional development
United States	5	8	Early and sustained engagement at the highest levels in the plotted sample
Argentina	2	5	Growth in human rights and governance-related programs

Interpretation of Scores

Score Range	Meaning
0	No engagement
1-2	Minimal involvement
3-4	Moderate engagement
5-6	Strong engagement
7-8	Extensive and institutionalized engagement

2. Cross-Country Regression Graphs

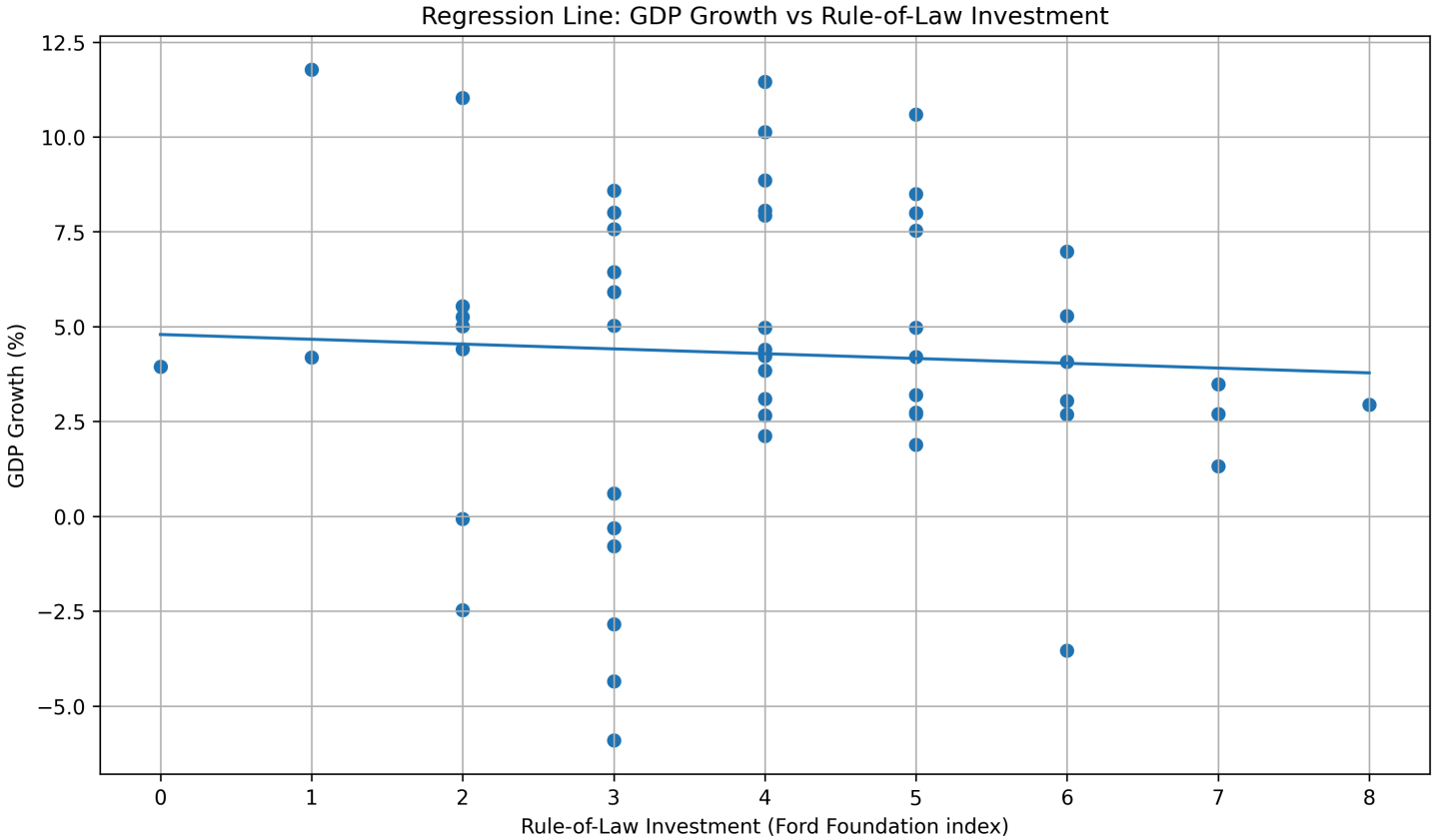
GDP Growth vs Rule-of-Law Investment Scatter



This scatter plot shows the raw relationship between the Ford Foundation rule-of-law investment index on the horizontal axis and GDP growth on the vertical axis. Each point is one country-year observation.

The main visual pattern is dispersion rather than a clear linear structure. High index values do not consistently correspond to high GDP growth, and mid-range index values can be associated with either low or high growth. The United States sits at the high end of the index series but remains in a moderate growth range, while countries such as China and India achieve much higher growth at lower index values. The figure therefore suggests that the unconditional relationship between the two variables is weak.

GDP Growth vs Rule-of-Law Investment Regression Line



This figure adds a fitted linear trend line to the same scatter relationship. The line is almost flat, which indicates that the average linear association between the plotted index measure and GDP growth is very small.

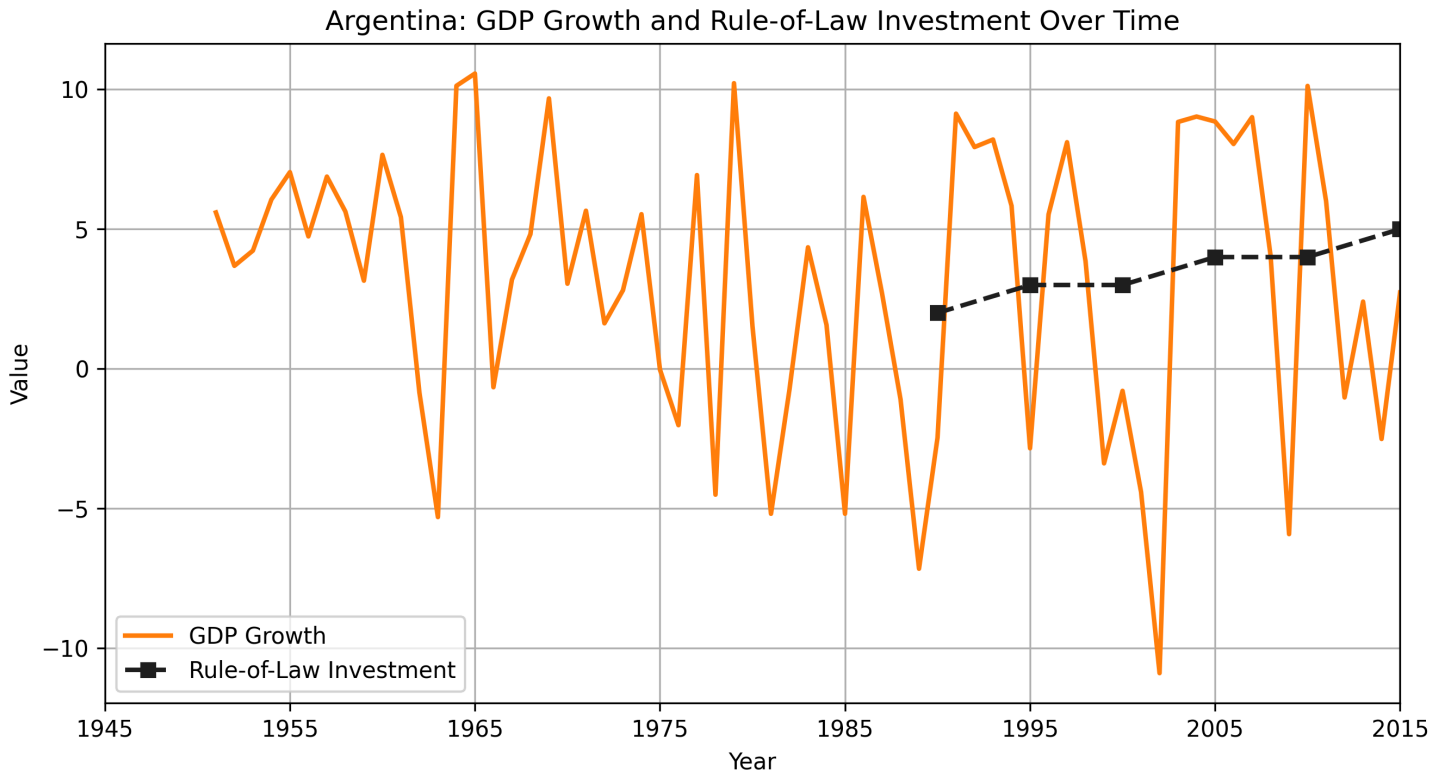
The wide spread of points around the line matters as much as the slope itself. Even where the fitted line rises slightly, the underlying observations remain highly scattered, so the visual evidence points to a weak aggregate relationship rather than a strong pattern.

Overall Visual Pattern

Across all figures, the rule-of-law investment index tends to increase smoothly over time, while GDP growth varies much more sharply both across countries and across years. The scatter plot and regression-line plot show that the average cross-country relationship is weak, and the country trend plots show that growth dynamics are highly country-specific rather than tightly linked to the index measure alone.

3. Country-by-Country Integrated Sections

Argentina



Argentina's graph shows a steadily rising rule-of-law investment index from 2 to 5, while GDP growth moves sharply across the period. Growth is negative in the early years, becomes positive later, and peaks strongly around 2005 to 2010 before easing again by 2015.

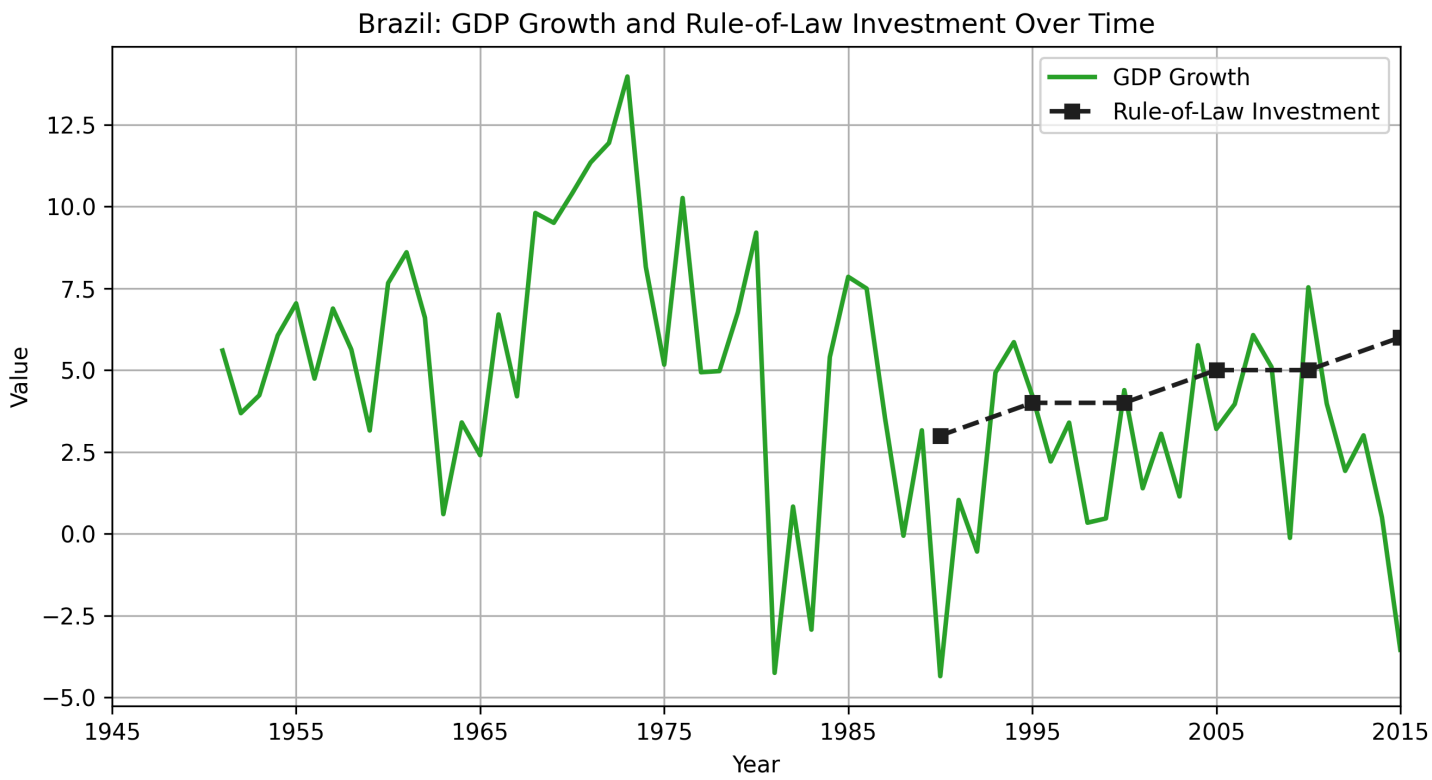
The image shows that the index series follows a smooth upward path, but GDP growth is cyclical and unstable. The two series do not move together in a simple one-to-one way.

Country-specific interpretation: growth in human rights programs, with index scores rising from 2 in 1990 to 5 in 2015.

Year	Index Score Used in Plot	Justification
1990	2	Continued low-level activity
1995	3	Expansion in human rights programs
2000	3	Moderate engagement
2005	4	Governance and legal reform

Year	Index Score Used in Plot	Justification
2010	4	Sustained engagement
2015	5	Increased institutional involvement

Brazil



Brazil also shows a steadily increasing investment index, rising from 3 to 6. GDP growth, however, is much more volatile. The series starts with negative growth in 1990, improves during the middle years, strengthens again around 2010, and then falls back into negative territory by 2015.

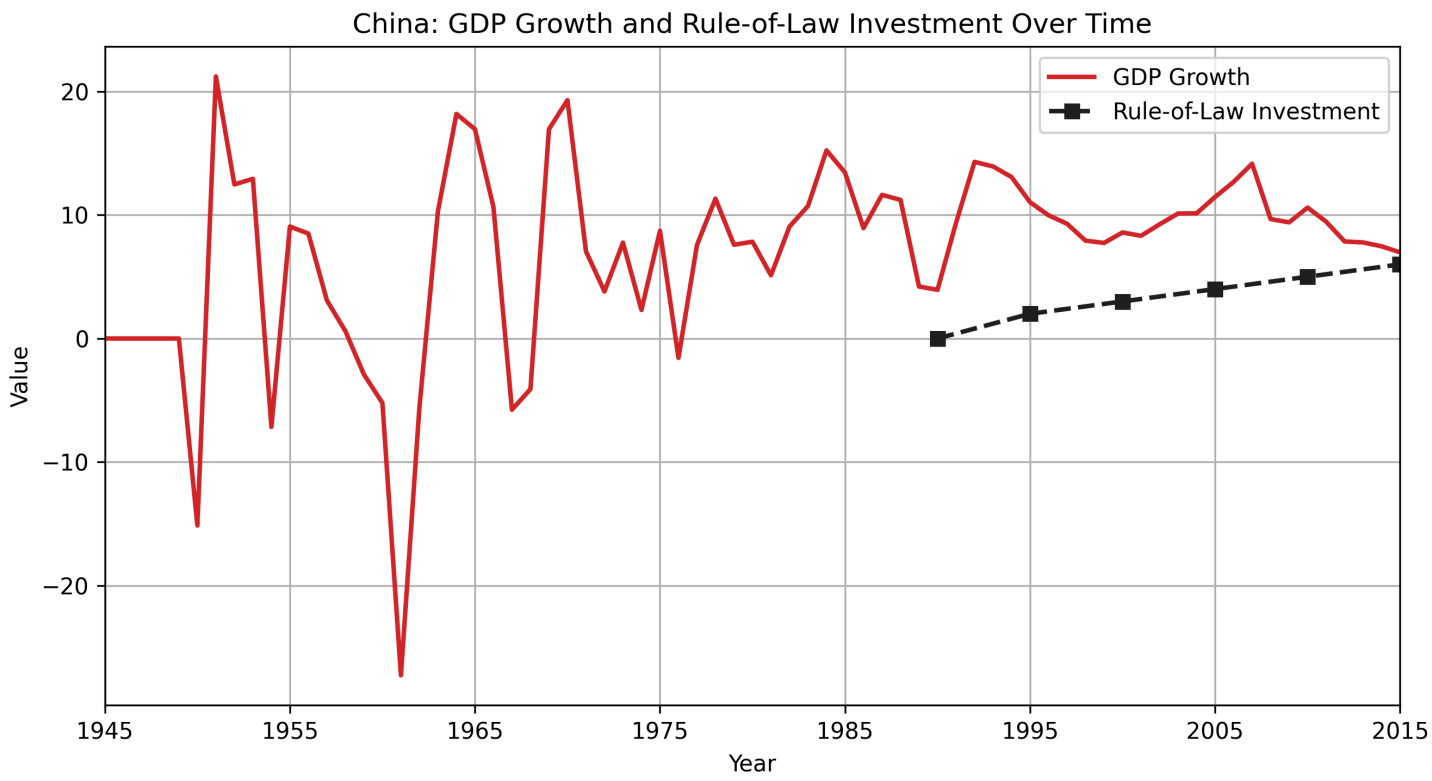
The figure highlights the contrast between a smooth increase in the index variable and strong macroeconomic fluctuations in growth.

Country-specific interpretation: expansion in governance and legal advocacy, with index scores rising from 3 in 1990 to 6 in 2015.

Year	Index Score Used in Plot	Justification
1990	3	Continued engagement
1995	4	Expansion in human rights and legal reform

Year	Index Score Used in Plot	Justification
2000	4	Multi-domain engagement
2005	5	Sustained institutional programs
2010	5	Continued high engagement
2015	6	Strong and broad involvement

China



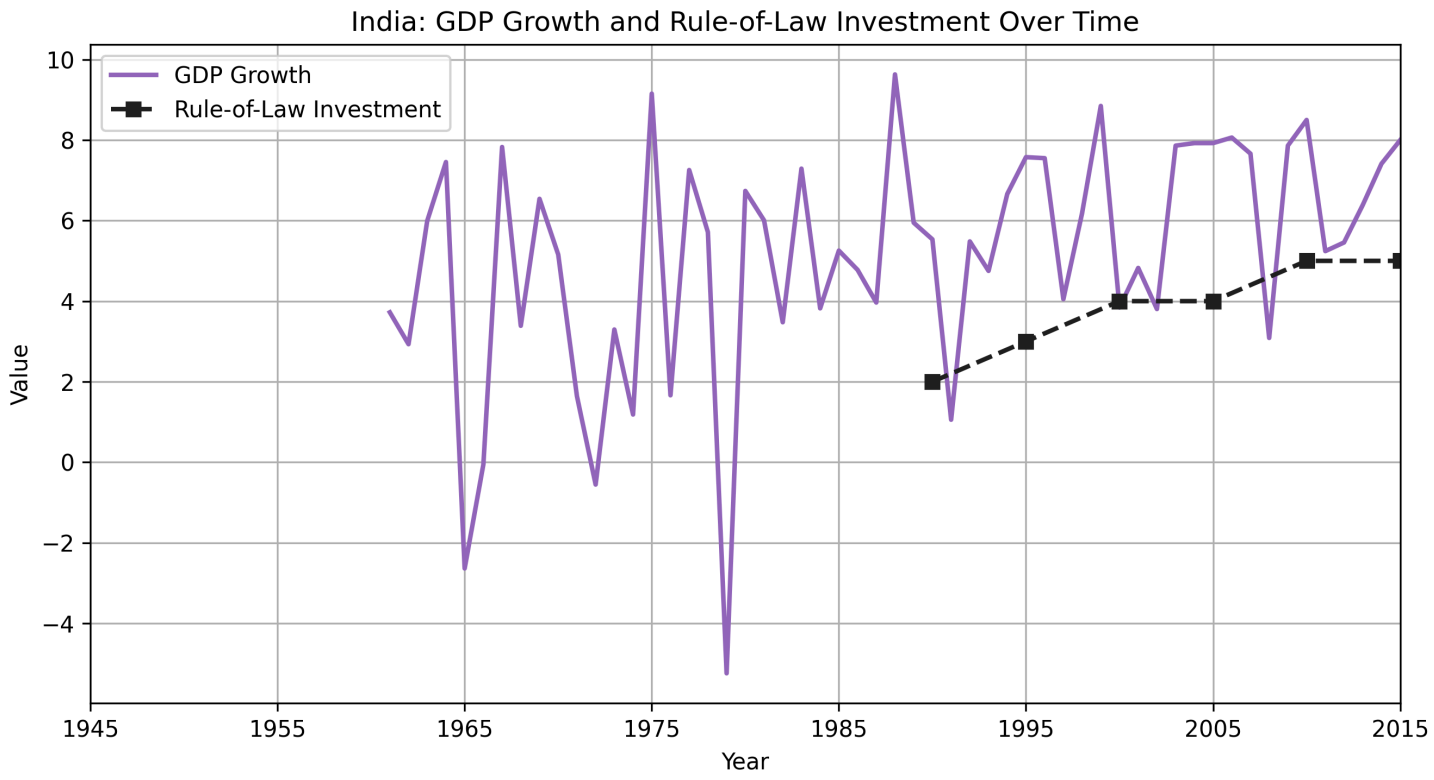
China's graph combines a rising index series with persistently high GDP growth. The index increases from 0 to 6, while GDP growth remains high throughout the period and reaches especially strong levels in the mid-1990s, mid-2000s, and around 2010.

Compared with the other countries, China stands out as the clearest high-growth case. The figure shows that strong growth can coexist with a moderate, gradually rising index value, but the magnitude of Chinese growth is much larger than the movement in the index itself.

Country-specific interpretation: no activity pre-1990 in the conceptual framework, followed by expansion after legal reforms, with index scores rising from 0 in 1990 to 6 in 2015.

Year	Index Score Used in Plot	Justification
1990	0	No documented legal or governance programs in the earlier conceptual framework
1995	2	Initial engagement through legal education programs
2000	3	Expansion into legal training and institutional support
2005	4	Broader engagement including legal clinics and governance reform
2010	5	Sustained multi-domain engagement
2015	6	Institutionalized and long-term involvement

India



India's figure shows a steady rise in the investment index from 2 to 5. GDP growth stays relatively high across the whole period, dips around 2000, and then returns to a strong range after 2005.

The image suggests broad long-run improvement in both series, but the timing is not perfectly aligned. GDP growth changes more abruptly than the index path.

Country-specific interpretation: continuous but moderate engagement, with index scores rising from 2 in 1990 to 5 in 2015.

Year	Index Score Used in Plot	Justification
1990	2	Continued but limited engagement
1995	3	Expansion into public interest law
2000	4	Multiple activities including governance and legal advocacy
2005	4	Sustained engagement
2010	5	Broader institutional involvement
2015	5	Continued high engagement

Kenya

Kenya shows a gradual increase in the index series from 1 to 5. GDP growth is positive at the beginning, weak around 2000, then rebounds strongly by 2005 and 2010 before easing somewhat by 2015.

The figure presents a medium-run improvement story, but again the movement in GDP growth is more uneven than the steady increase in the index variable.

Country-specific interpretation: increasing engagement over time, with index scores rising from 1 in 1990 to 5 in 2015.

Year	Index Score Used in Plot	Justification
1990	1	Limited engagement
1995	2	Initial expansion
2000	3	Access to justice programs emerge
2005	3	Continued engagement
2010	4	Broader governance involvement
2015	5	Sustained multi-domain engagement

Mexico

Mexico's graph is marked by a very large drop in GDP growth in 1995, followed by recovery and continued fluctuation afterward. Over the same period, the index rises smoothly from 2 to 5.

The visual contrast is strong: the index variable increases in a stable way, while GDP growth is highly sensitive to short-run shocks. The large 1995 downturn dominates the pattern.

Country-specific interpretation: gradual increase in institutional reform, with index scores rising from 2 in 1990 to 5 in 2015.

Year	Index Score Used in Plot	Justification
1990	2	Small-scale programs
1995	3	Expansion into justice system reform
2000	3	Moderate engagement
2005	4	Broader governance programs
2010	4	Sustained engagement
2015	5	Increased institutional involvement

Nigeria

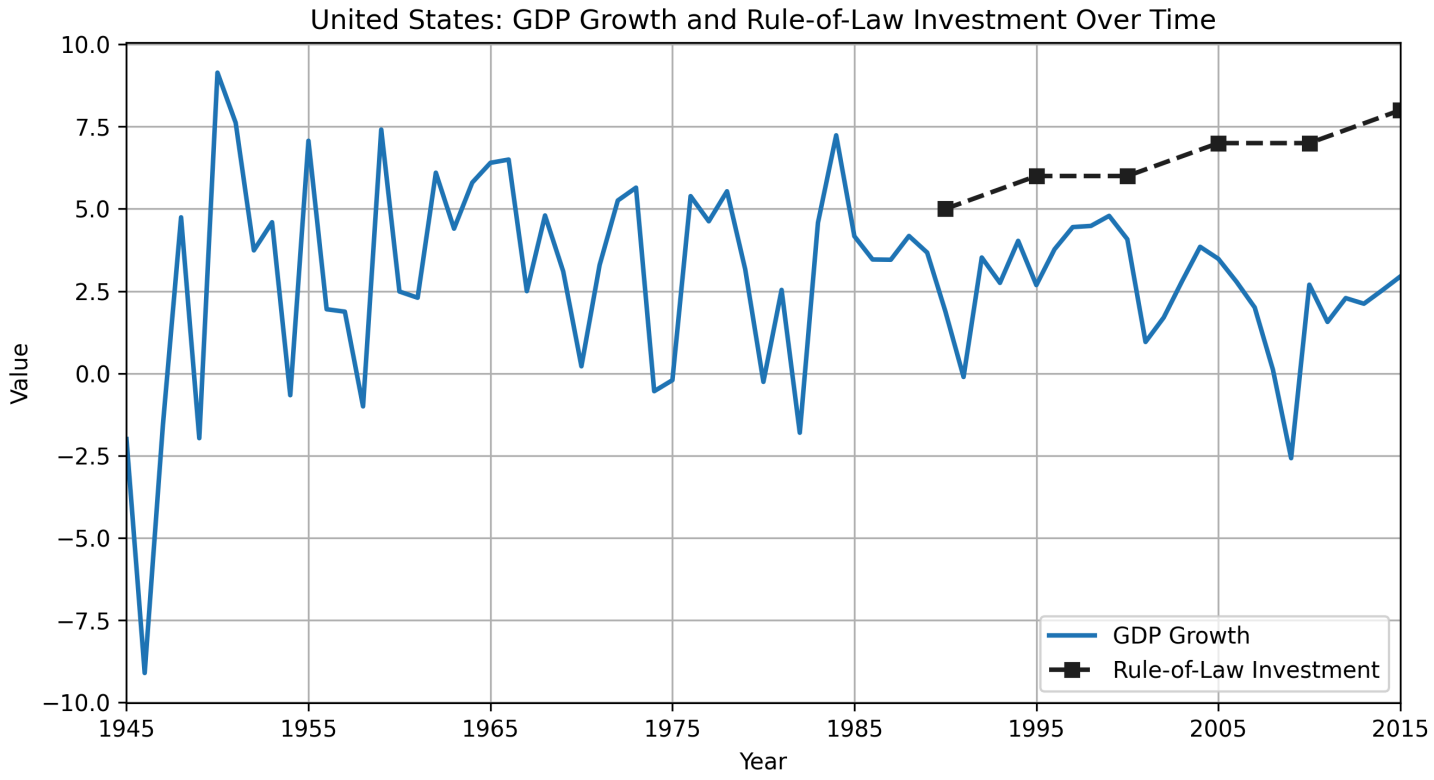
Nigeria's figure shows one of the most volatile GDP growth paths in the sample. The index rises from 1 to 4, but GDP growth begins very high in 1990, drops close to zero by 1995, then recovers and remains positive before slowing again by 2015.

The graph makes clear that GDP growth fluctuates much more sharply than the index series. The two series move on very different scales and with very different levels of stability.

Country-specific interpretation: limited and less continuous engagement, with index scores rising from 1 in 1990 to 4 in 2015.

Year	Index Score Used in Plot	Justification
1990	1	Limited activities
1995	2	Small-scale programs
2000	2	Continued limited engagement
2005	3	Some expansion
2010	3	Moderate involvement
2015	4	Increased engagement

United States



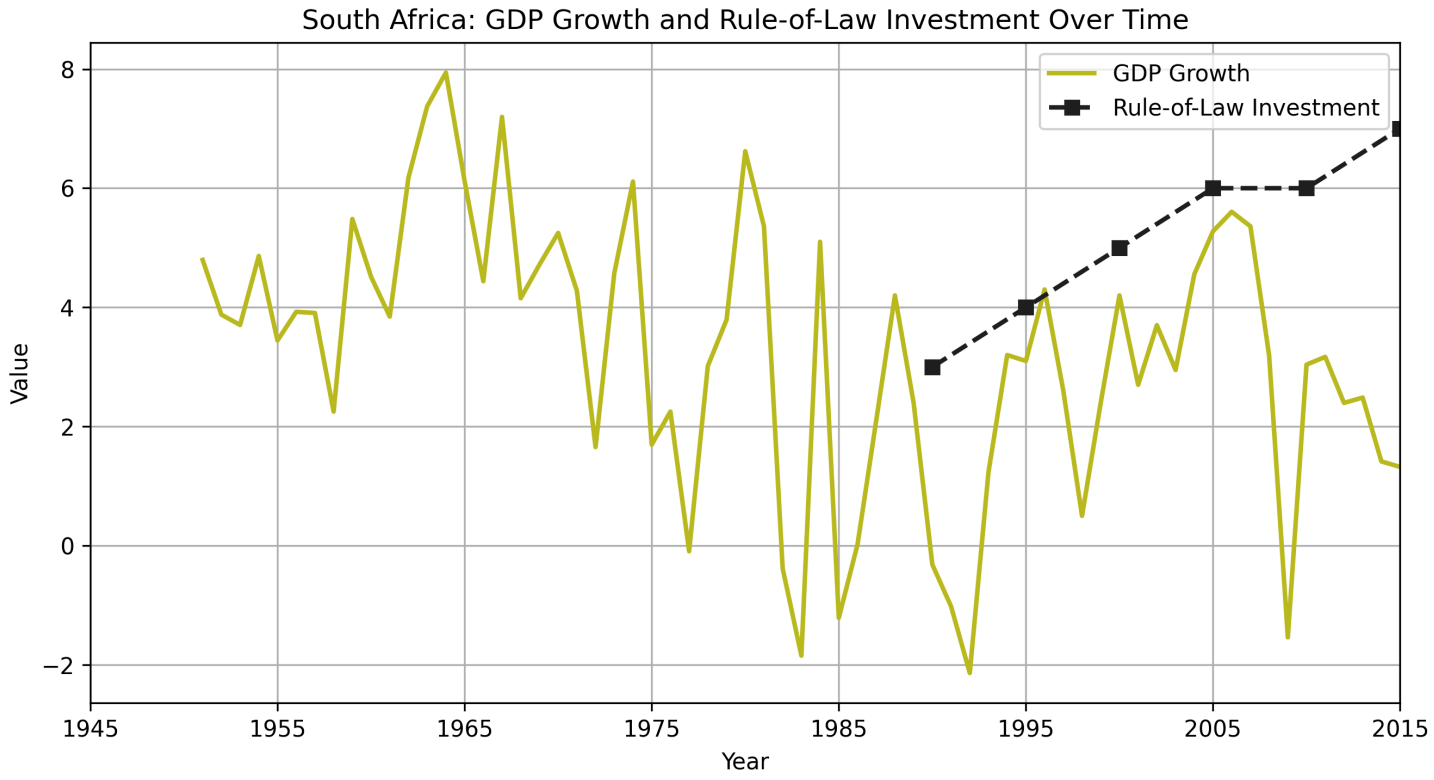
The United States has the highest rule-of-law investment index values in the sample, increasing from 5 to 8. GDP growth remains comparatively stable and stays mostly in the 2% to 4% range.

This figure differs from the high-growth emerging-market cases. It shows high index values combined with moderate, low-volatility growth rather than rapid expansion.

Country-specific interpretation: early and sustained engagement, with index scores rising from 5 in 1990 to 8 in 2015.

Year	Index Score Used in Plot	Justification
1990	5	Continued high engagement
1995	6	Expanded legal and governance initiatives
2000	6	Sustained involvement
2005	7	Broad institutional engagement
2010	7	Continued high-level activity
2015	8	Long-term institutionalized presence

South Africa



South Africa's graph shows the index series rising from 3 to 7. GDP growth starts low, improves through the middle of the sample, peaks around 2005, and then weakens again by 2015.

The pattern is similar to several other countries in the sample: the index rises steadily, but GDP growth follows a more cyclical path with both recovery and slowdown phases.

Country-specific interpretation: strong post-1990 institutional development, with index scores rising from 3 in 1990 to 7 in 2015.

Year	Index Score Used in Plot	Justification
1990	3	Political transition increases activity
1995	4	Expansion in legal reform
2000	5	Strong institutional involvement
2005	6	Sustained high engagement
2010	6	Continued programs
2015	7	Deep institutionalization

4. Robustness, Alternative Specifications, and Interpretation

Robustness and Limitations

Issue	Explanation
Measurement uncertainty	Based on qualitative sources
Ordinal scale	Not proportional differences
Coding subjectivity	Interpretation-dependent
Time consistency	Uneven data availability
External validity	Reflects Ford Foundation only

Several robustness considerations apply to the constructed rule-of-law index:

1. Measurement Uncertainty

The index is based on qualitative reports rather than standardized quantitative data. As a result, it may introduce measurement error due to subjective interpretation.

2. Ordinal Nature

The index is ordinal rather than continuous. Differences between values, such as 2 versus 4, do not represent proportional differences in actual investment.

3. Alternative Interpretations

To test robustness, alternative specifications can be considered:

- Binary indicator (presence vs absence of rule-of-law programs)
- Categorical grouping (low, medium, high engagement)
- Lagged index (to account for delayed effects on economic growth)

Approach	Description
Binary	Presence vs absence
Categorical	Low / Medium / High
Lagged index	Time-delayed effects

4. Time Consistency

The index accounts for historical timing, but uneven data availability across countries may affect comparability.

5. External Validity

The index reflects Ford Foundation engagement specifically and may not capture broader rule-of-law conditions measured by other indices, such as the World Bank or World Justice Project.

Key Interpretation

- Scores are constant within phases, not independently observed yearly.
- Yearly values reflect period-based coding applied consistently.

- Changes only occur when new activities appear, scope expands, or engagement becomes sustained.

Important Note

The yearly scores should not be interpreted as precise annual measurements. They represent structured approximations of engagement intensity based on qualitative evidence.

Key Statement

The rule-of-law index is a constructed, unitless, ordinal measure reflecting the relative intensity of Ford Foundation engagement based on observable activities, rather than actual financial investment or standardized quantitative data.